

Q1: If all the links in the Internet were to provide reliable delivery service, would the TCP reliable delivery service be redundant? Give 2 reasons to support your answer. [4]

Yes/No? _____

Reason 1:

Reason 2:

Q2: Suppose two nodes start to transmit at the same time a packet of length L over a broadcast channel of rate R . Denote the propagation delay between the two nodes as d_{prop} . Will there be a collision if $d_{prop} > L/R$? Give reason to support your answer. [4]

Yes/No? _____

Reason:

Q3: In CSMA/CD, after the fifth collision, what are the possible values that a node can take to delay its transmission? [2]

Q4: In Binary exponential back off algorithm for CSMA/CD, after the 11th collision, what are the possible values that a node can take to delay its transmission? [2]

National University of Computer and Emerging Sciences, Lahore Campus



Course: Computer Networks
Program: BS(Computer Science)
Duration: 20 Minutes
Date: 27 Nov, 2019
Section: E

Course Code: CS307
Semester: Fall 2019
Total Marks: 20
Quiz: 4
Page(s): 2

Name _____ Roll No. _____

Question 1: [Marks 12]

For each of the following IP address ranges, specify the network address, broadcast address, and maximum number of host IPs available. (Show all your work).

- **192.168.100.0/24**

N/W: 192.168.100.0
B/C: 192.168.100.255
No. of IPs: 254

- **100.10.8.0/22**

N/W: 100.10.8.0
B/C: 100.10.11.255
No. of IPs: 1022

- **202.1.0.0/16**

N/W: 202.1.0.0
B/C: 202.1.255.255
No. of IPs: 65534

- **101.51.192.0/18**

N/W: 101.51.192.0
B/C: 101.51.255.255
No. of IPs: 16382

Question 2: [Marks 8]

What is fragmentation. How does the IP layer specify that a packet is fragmented (Name the IP header fields and their applications).

Fragmentation is done by the network layer when the maximum size of datagram is greater than maximum size of data that can be held a frame i.e., its Maximum Transmission Unit (MTU). The network layer divides the datagram received from transport layer into fragments so that data flow is not disrupted.

Fields in IP header for fragmentation

- Fragment offset (13 bits) – use to identify sequence of fragments in the frame.
- More fragments (MF = 1 bit) – tells if more fragments ahead of this fragment.
- Don't fragment (DF = 1 bit) – if we don't want the packet to be fragmented then DF is set to 1.

National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Computer Networks	Course Code:	CS 3001
	Program:	BS (Computer Science)	Semester:	Fall 2023
	Duration:	15 minutes	Total Marks:	15
	Paper Date:	02-November-2023	Section	5J
	Exam Type:	Quiz 4 - Chapter 4	Page(s):	2

Student Name

Roll No.

Section:

Q1. Encircle the correct option:

[5 marks] [CLO 3]

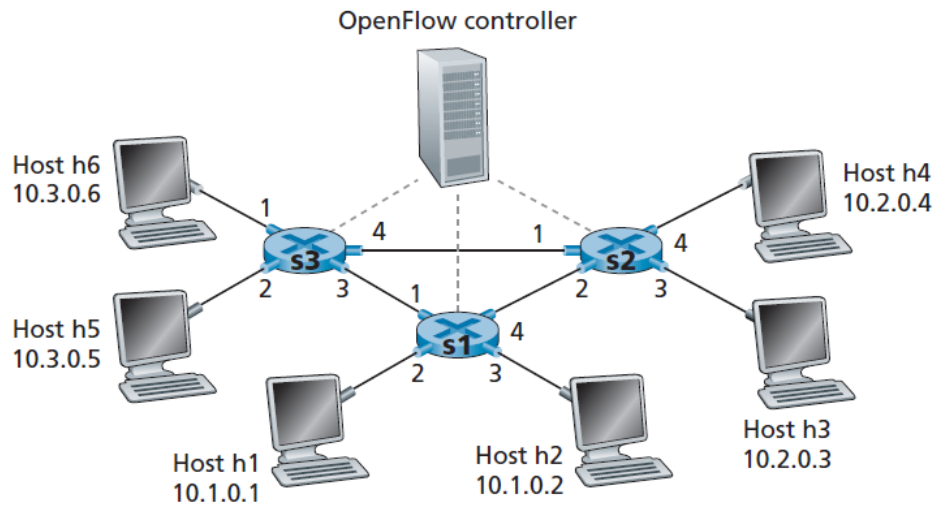
1. The key function of the Network Layer's data plane is
 - a) Error and Diagnostic Functions
 - b) Routing Algorithms
 - c) Routing / network wide logic
 - d) Forwarding / local per router function

2. Match the classes of IP addresses with their functional applications
 - Class A ----- i) Multicast services
 - Class B ----- ii) Reserved for future use
 - Class D ----- iii) Large organizations
 - Class E ----- iv) Midsize Organizations.
 - a) Class A — iv, Class B — iii, Class D — ii, Class E — i
 - b) Class A — ii, Class B — iv, Class D — i, Class E — iii
 - c) Class A — iii, Class B — iv, Class D — i, Class E — ii
 - d) Class A — i, Class B — ii, Class C — iii, Class D — iv

3. Network ID and a subnet ID
 - a) can never be the same
 - b) can be same
 - c) are always the same.
 - d) None of the above

True/False:

1. The value in the header length field of IPv4 header is scaled by a factor of 2. [T / F]
2. The IP version 4 address consist of 4 octets. [T / F]



Q2. Consider again the SDN OpenFlow network shown above. Suppose that the desired forwarding behavior for datagrams arriving from hosts h3 or h4 at s2 is as follows:

- any datagrams arriving from host h3 and destined for h1, h2, h5 or h6 should be forwarded in a clockwise direction in the network
- any datagrams arriving from host h4 and destined for h1, h2, h5 or h6 should be forwarded in a counter-clockwise direction in the network.

Specify the flow table entries in s2 that implement this forwarding behavior.

[10 marks] [CLO 3]

S2 Flow Table

Match	Action

Solution Quiz#4(5J)

MCQ's

1. d
2. c
3. b

T/F

1. F
2. T

Q2

S2 Flow Table	
Match	Action
Ingress Port = 3; IP Dst = 10.1.*.* Ingress Port = 3; IP Dst = 10.3.*.*	Forward (2) Forward (2)
Ingress Port = 4; IP Dst = 10.1.*.* Ingress Port = 4; IP Dst = 10.3.*.*	Forward (1) Forward (1)

National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Computer Networks	Course Code:	CS 3001
	Program:	BS (Computer Science)	Semester:	Fall 2023
	Duration:	15 minutes	Total Marks:	15
	Paper Date:	02-November-2023	Section	5K
	Exam Type:	Quiz 4 - Chapter 4	Page(s):	2

Student Name

Roll No.

Section:

Q1. Encircle the correct option:

[5 marks] [CLO 3]

1. The key function of the Network Layer's control plane is
 - a) Error and Diagnostic Functions
 - b) Routing Algorithms
 - c) Forwarding / local per router function
 - d) Routing / network wide logic

2. Match the classes of IP addresses with their functional applications
 - Class A ----- i) Multicast services
 - Class B ----- ii) Reserved for future use
 - Class D ----- iii) Large organizations
 - Class E ----- iv) Midsize Organizations.
 - a) Class A — iv, Class B — iii, Class D — ii, Class E — i
 - b) Class A — ii, Class B — iv, Class D — i, Class E — iii
 - c) Class A — iii, Class B — iv, Class D — i, Class E — ii
 - d) Class A — i, Class B — ii, Class C — iii, Class D — iv

3. The preferred approach taken for the transitioning from IPv4 to IPv6 is
 - a) Flag days
 - b) tunneling
 - c) dual stack approach
 - d) Transitioning not started

True/False:

1. VLSM is possible with both Classful and Classless IPv4 addressing schemes. [T / F]
2. The IP version 6 address consists of 16 octets. [T / F]

Q2. Consider a subnet with prefix 128.119.40.128/26. Give an example of one IP address (of form xxx.xxx.xxx.xxx) that can be assigned to this network. Suppose an ISP owns the block of addresses of the form 128.119.40.64/26. Suppose it wants to create four subnets from this block, with each block having the same number of IP addresses. What are the prefixes (of form a.b.c.d/x) for the four subnets? (**Note:** All answers should be in dotted decimal notation.)

[10 marks] [CLO 3]

Solution Quiz#4(5K)

MCQ's

1. d
2. c
3. b

T/F

1. F
2. T

Q2

Any IP address in range 128.119.40.128 to 128.119.40.191

Four equal size subnets: 128.119.40.64/28, 128.119.40.80/28, 128.119.40.96/28, 128.119.40.112/28

(25)

National University of Computer and Emerging Sciences, Lahore
Campus Quiz4 [BCS: Section 5A] Fall 2024

Computer Networks (Code: CS3001)

Quiz Date: October 31, 2024

Total Marks: 20

Duration: 20 -Minutes

Name _____

-- Roll #-----

Instructions: Attempt all questions on this sheet. You can make use of a rough sheet (do not attach to this sheet). Cutting/Overwriting will be considered incorrect.

Q1: For each of the following IP address ranges, specify the network address, broadcast address, and maximum number of host IPs available at the space provided in the Table. [12 Marks] (CLO 3)

assumption:
usable host
↑ IPs

CIDR	Network Address	Broadcast Address	Maximum Hosts
192.168.100.0/24	192.168.100.0/24 ✓	192.168.100.255/24 ✓	$2^8 = 256 - 2 = 254$
100.10.8.0/22	100.10.8.0/22 ✓	100.10.11.255/22 ✓	$2^{10} = 1024 - 2 = 1022$
202.1.0.0/16	202.1.0.0/16 ✓	202.1.255.255/16 ✓	$2^{16} = 65536 - 2 = 65534$
101.51.192.0/18	101.51.192.0/18 ✓	101.51.255.255/18 ✓	$2^{14} = 16384 - 2 = 16382$

Q2: An organization is granted a block of addresses starting with 132.100.24.0/23 (512 addresses). The organization needs to have four sub-blocks of addresses to use in its four subnets. Sub-blocks are designed in such a way that 1st one can accommodate 256 addresses, 2nd can accommodate 128 addresses and remaining two sub-blocks can accommodate 64 addresses each (including network address and broadcast address). With reference to this scenario, answer the following: [8 Marks] (CLO 3)

A. Write the subnet mask for each sub-block.

Subnet mask for the 1st sub-block:

255.255.255.0 ✓

Subnet mask for the 2nd sub-block:

255.255.255.128 ✓

Subnet mask for the 3rd sub-block:

255.255.255.192 ✓

Subnet mask for the 4th sub-block:

255.255.255.192 ✓

B. Write the 25th and 50th host address for 1st and 3rd sub-blocks.

25th host address for 1st sub-block:

132.100.24.25 ✓

50th host address for 1st sub-block:

132.100.24.50 ✓

25th host address for 3rd sub-block:

132.100.25.153 ✓

50th host address for 3rd sub-block:

132.100.25.178 ✓

132.100.24.0/23

10000100.01100100.00011000.00000000

A: 132.100.24.0 (network address)

C: 132.100.25.128 (network address)

A: 256 → 0

B: 128 → 1 → 10

C: 64

11 → 110

D: 64

111

Computer Networks (CS3001)

Date: December 03, 2024

Course Instructor

Ms. Saba Tariq

Quiz 4(b)

Total Time (min): 20

Total Marks: 20

Total Questions: 2

Solution

Roll No

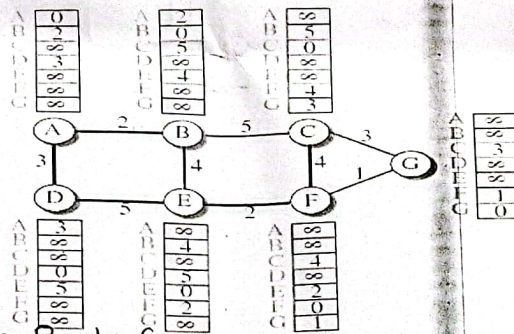
Section

Student Signature

Do not write below this line

Attempt all the questions.

Q1: Update the distance vectors of C and D router in first iteration ($t=1$) for the following network.



Updating Router C

$$A: 5 + 2 = 7$$

$$B: 5$$

$$C: 0$$

$$D: \infty$$

$$E: 6$$

$$F: 4$$

$$G: 3$$

Updating router D

$$A: 3$$

$$B: 5$$

$$C: \infty$$

$$D: 0$$

$$E: 5$$

$$F: 7$$

$$G: 2$$

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Q2: Update the Link State Database for C and D router in above network using Dijkstra Algorithm.

Router C:

	A	B	C	D	E	F	G
C	∞	∞	0	∞	∞	∞	∞
G	∞	5		∞	∞	4	3
F	∞	5		∞	∞	4	
B	∞	5		∞	6		
E	7			∞	6		
A	7			11			
				10			

Router D:

	A	B	C	D	E	F	G
D	∞	∞	∞	0	∞	∞	∞
A	3	∞	∞		5	∞	∞
B		5	∞		5	∞	∞
E			10		5	∞	∞
F			10			7	∞
G			10				8
			10				

Computer Networks (CS3001)

Date: December 03, 2024

Course Instructor

Ms. Saba Tariq

Quiz 4(a)

Total Time (min): 20

Total Marks: 20

Total Questions: 2

Solution

Roll No

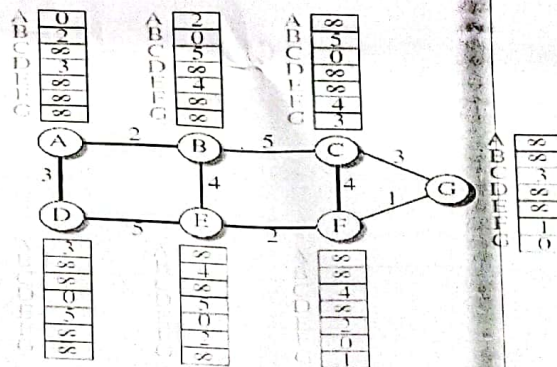
Section

Student Signature

Do not write below this line

Attempt all the questions.

Q1: Update the distance vectors of A and F router in first iteration ($t=1$) for the following network.



Router A:

A: 0

B: 2

C: 7

D: 3

E: 6

F: ∞

G: ∞

Router F:

A: ∞

B: 6

C: 4

D: 7

E: 2

F: 0

G: 1

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Lahore Campus

Q2: Update the Link State Database for A and F router in above network using Dijkstra Algorithm.

Router A:

	A	B	C	D	E	F	G
A	0	∞	∞	∞	∞	∞	∞
B		2	∞	3	∞	∞	∞
D			7	3	6	∞	∞
E			7		6	∞	∞
C			7			8	∞
F						8	10
G							9

Router F:

	A	B	C	D	E	F	G
F	∞	∞	∞	∞	∞	0	∞
G	∞	∞	4	∞	2		1
E	∞	∞	4	∞	2		
C	∞	6	4	7			
B	∞	6		7			
D	8			7			
A	8						

**National University of Computer and Emerging Sciences,
Lahore Campus**

	Course Name:	Computer Networks	Course Code:	CS 3001
	Program:	BS (Computer Science)	Semester:	Fall 2025
	Duration:	15 minutes	Total Marks:	15
	Paper Date:	21-October-2025	Section	5A , 5C
	Exam Type:	Quiz 4 - Chapter 4	Page(s):	2

Student Name

Roll No.

Section:

Q1. Encircle the correct option:

[5 marks] [CLO 3]

- What is the purpose of a Subnet mask in IP addressing??
 - To identify the Subnet portion of an IP address
 - To encrypt IP traffic
 - To define the Default gateway
 - To allocate Dynamic IP Addresses
- Which TCP/IP model layer does DHCP work at?
 - Internet
 - Session
 - Application
 - Transport
- An IP layer datagram has arrived with the first 8 bits as 0110 0110. Which of the following is correct?
 - IPv4, Header Length: 80 bytes
 - IPv4, Header Length: 32 bytes
 - IPv6, Some Priority / Traffic Class
 - IPv6, Header Length: 24 bytes

True/False:

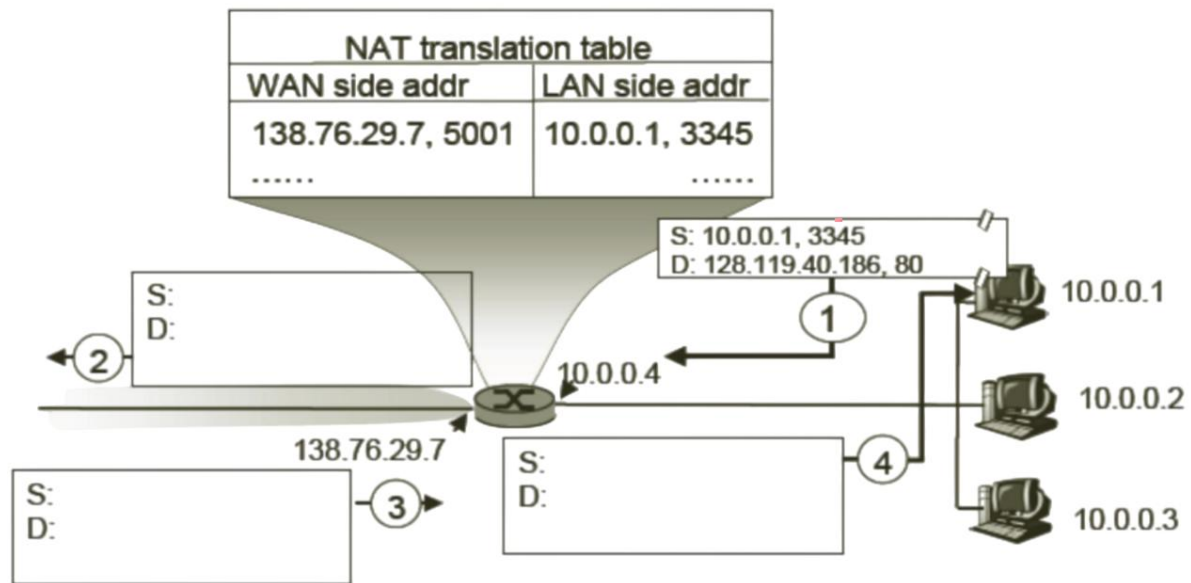
- DHCP can provide not only IP addresses but also subnet masks, gateways, and DNS server information. [T / F]
- Class A IP addresses have the first octet (8-bits) reserved for the network portion, and the remaining 3 octets for hosts. [T / F]

Q2:

[6 Marks] [CLO 3]

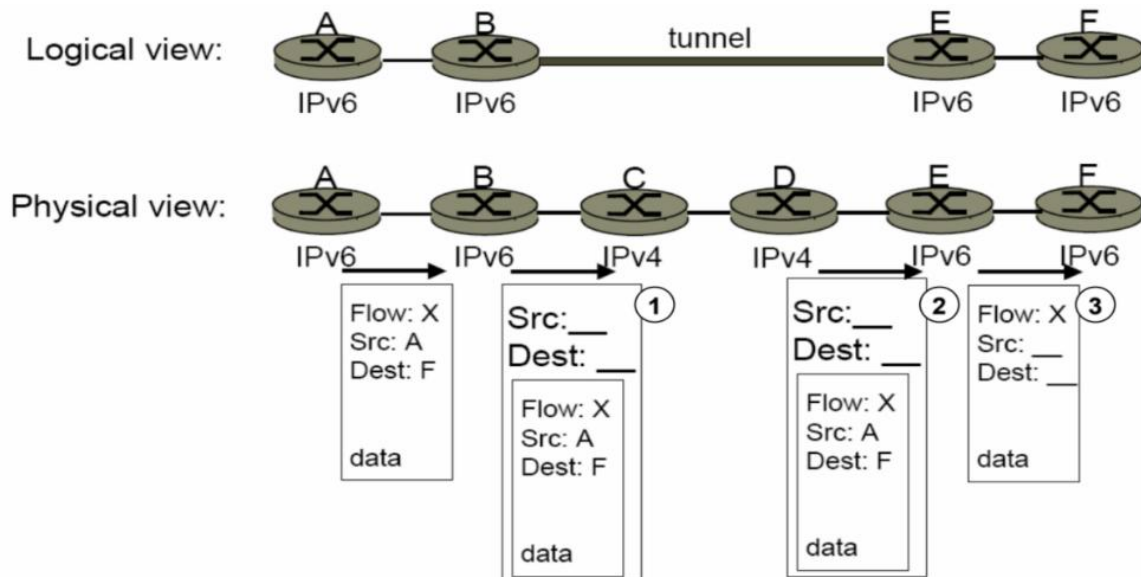
The diagram below shows a packet traveling through a NAT router. Packet 1 is sent from the internal host (S) to the NAT router, packet 2 is sent from the NAT router to the external web server (D), packet 3 is received from the web server by the NAT router, and packet 4 is sent by the NAT router to the original host. Fill in the missing source and destination IP

addresses and port numbers in packets 2-4 (ie fill in the S and D values in the given Boxes 2, 3 and 4):



Q3: [3 Marks] [CLO 3]

The diagram below shows an IPv6 packet tunneled over IPv4 across routers with IP addresses A, B, C, D, E & F. Fill in the missing source and destination addresses at places/packets marked 1, 2, and 3 (fill in the diagram).



Q4: [1 Mark] [CLO 3]

We know that an IPv4 Address is 32 bits long. We also know that a bit can either be a 1 or 0. How many total IPv4 Addresses can possibly exist?

	Course Name:	Computer Networks	Course Code:	CS 3001
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	Exam Type:	Quiz 4 - Chapter 4	Page(s):	2

Student Name

Roll No.

Section:

Q1. Encircle the correct option:

[5 marks] [CLO 3]

- An IP packet has arrived with the first 8 bits as 0100 0110. Which of the following is correct?
 - IPv4, Header Length: 24 bytes
 - IPv4, Header Length: 32 bytes
 - IPv6, Header Length: 24 bytes
 - IPv6, Header Length: 32 bytes
- In Classless Inter-Domain Routing (CIDR), the network ID in an IPv4 address is:
 - Always 8 bits long
 - Always 16 bits long
 - Always 24 bits long
 - None of the above
- Which IP address(es) should never be assignable to host or router interfaces?
 - Network Address
 - Broadcast Address
 - Both a and b
 - None of the above

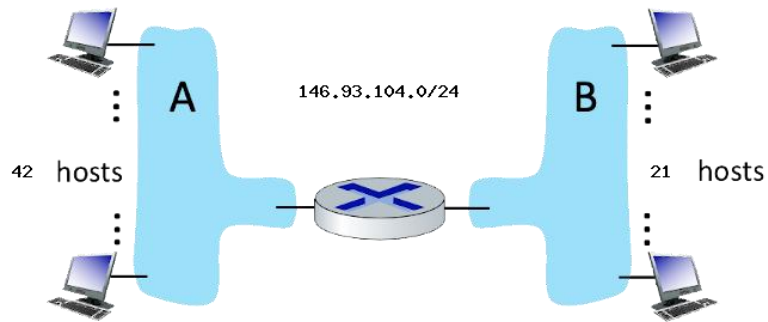
True/False:

- A DHCP client that has been assigned an Address for a period can release the Address before the Expiration time. [T / F]
- VLSM is possible with both Classful and Classless IPv4 Addressing Schemes. [T / F]

Q2:

[2+1+1+1+1+1+1+1+1 = 10 Marks] [CLO 3]

Consider the router and the two attached subnets below (A and B). The number of hosts is also shown below. The subnets share the 24 high-order bits of the address space: 146.93.104.0/24. Assign subnet addresses to each of the subnets (A and B) so that the amount of address space assigned is minimal, and at the same time leaving the largest possible contiguous address space available for assignment if a new subnet were to be added in the future. Then answer the questions below:



- (1) How many **Hosts** can there be in this Address Space?
- (2) What is the **Subnet Address** of Subnet A? (**CIDR notation**)
- (3) What is the **Broadcast Address** of Subnet A?
- (4) What is the **First Usable Address** of Subnet A?
- (5) What is the **Last Usable Address** of Subnet A?
- (6) What is the **Subnet Address** of Subnet B? (**CIDR notation**)
- (7) What is the **Broadcast Address** of Subnet B?
- (8) What is the **First Usable Address** of Subnet B?
- (9) What is the **Last Usable Address** of Subnet B?

SOLUTION QUIZ-04 BCS-5A

QUESTION-01

1. A
2. C
3. C

TRUE/FALSE

- i. True
- ii. True

QUESTION-02

Packet 2: S=138.76.29.7, 5001 D=128.119.40.186, 80

Packet 3: S=128.119.40.186, 80 D=138.76.29.7, 5001

Packet 4: S=128.119.40.186, 80 D=10.0.0.1, 3345

QUESTION-03

1: S=B D=E

2: S=B D=E

3: S=A D=F

QUESTION-04

$2^{32} = 4,294,967,296$ total IPv4 Addresses can exist

SOLUTION QUIZ-04 BCS-5C

QUESTION-01

1. A
2. D
3. C

TRUE/FALSE

- i. True
- ii. False

QUESTION-02

Sequential Assignment starting from the lower end of /24

1. 254
2. 146.93.104.0/26
3. 146.93.104.63
4. 146.93.104.1
5. 146.93.104.62
6. 146.93.104.64/27
7. 146.93.104.95
8. 146.93.104.65
9. 146.93.104.94

Sequential Assignment starting from the upper end of /24

1. 254
2. 146.93.104.192/26
3. 146.93.104.255
4. 146.93.104.193
5. 146.93.104.254
6. 146.93.104.160/27
7. 146.93.104.191
8. 146.93.104.161
9. 146.93.104.190

Solutions:

MCQs:

Multiple Choice Questions (MCQs):

1. **Answer: b**
2. **Answer: c**
3. **Answer: b**

True/False Questions:

- I. **Answer: True (T)**
- II. **Answer: True (T)**

QUESTION-02

We apply the supernet mask to see if we can find the beginning address.

205.16.37.44 AND 255.255.248.0 → 205.16.32.0

205.16.42.56 AND 255.255.248.0 → 205.16.40.0

205.17.33.76 AND 255.255.248.0 → 205.17.32.0

Only the first address belongs to this supernet.

**National University of Computer and Emerging Sciences, Lahore
Campus**

	Course Name:	Computer Networks	Course Code:	CS 3001
	Program:	BS (Software Engineering)	Semester:	Spring 2025
	Duration:	15 minutes	Total Marks:	15
	Paper Date:	08-May-2025	Section	6A
	Exam Type:	Quiz 4 - Chapter 4	Page(s):	2

Student Name

Roll No.

Section:

Q1. Encircle the correct option:

[5 marks] [CLO 3]

1. Which of the following best describes DHCP (Dynamic Host Configuration Protocol)?

- a) It encrypts packets for secure transmission.
- b) It dynamically assigns IP addresses to hosts on a network.
- c) It manages packet routing between autonomous systems.
- d) It detects packet transmission errors at routers.

2. Network Address Translation (NAT) is primarily used for:

- a) Converting IPv4 addresses to IPv6 addresses.
- b) Managing domain names.
- c) Allowing multiple devices to share a single public IP address.
- d) Detecting corrupted packets in the data link layer.

3. Subnetting helps to:

- a) Increase the speed of physical data transmission.
- b) Reduce network congestion by segmenting larger networks into smaller subnetworks.
- c) Guarantee end-to-end reliability.
- d) Encrypt sensitive network data.

True/False Questions:

I. DHCP can allocate IP addresses dynamically, automatically, or manually, depending on the network administrator's settings. [T/F]

II. NAT modifies the source or destination IP addresses of packets as they pass through a router. [T/F]

Question 2:**[Marks 10] [CLO 3]**

A supernet has a first address of 205.16.32.0 and a supernet mask of 255.255.248.0. A router receives three packets with the following destination addresses:

205.16.37.44

205.16.42.56

205.17.33.76

Which packet belongs to the supernet?

SOLUTION QUIZ-03 BCS-6C

QUESTION-01

MCQs

1. D
2. Class A – Large Organisations : Class B – Midsize Organisations : Class D – Multicast : Class E – Future Use
3. B

T/F

1. F
2. T

QUESTION-02

Any IP address in range 128.119.40.128 to 128.119.40.191

Four equal size subnets: 128.119.40.64/28, 128.119.40.80/28, 128.119.40.96/28, 128.119.40.112/28

**National University of Computer and Emerging Sciences, Lahore
Campus**

	Course Name:	Computer Networks	Course Code:	CS 3001
	Program:	BS (Software Engineering)	Semester:	Spring 2025
	Duration:	15 minutes	Total Marks:	15
	Paper Date:	08-April-2025	Section	6C
	Exam Type:	Quiz 4 - Chapter 4	Page(s):	2

Student Name

Roll No.

Section:

Q1. Encircle the correct option:

[5 marks] [CLO 3]

1. The key function of the Network Layer's data plane is
 - a) Error and Diagnostic Functions
 - b) Routing Algorithms
 - c) Routing / network wide logic
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2. Match the classes of IP addresses with their functional applications
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3. The preferred approach taken for the transitioning from IPv4 to IPv6 is
 - a) Flag days
 - b) tunneling
 - c) dual stack approach
 - d) Transitioning not started

True/False:

- I. The value in the header length field of IPv4 header is scaled by a factor of 2. [T / F]
- II. The IP version 4 address consist of 4 octets. [T / F]

Question 2:**[Marks 10] [CLO 3]**

Consider a subnet with prefix 128.119.40.128/26. Give an example of one IP address (of form xxx.xxx.xxx.xxx) that can be assigned to this network. Suppose an ISP owns the block of addresses of the form 128.119.40.64/26. Suppose it wants to create four subnets from this block, with each block having the same number of IP addresses. What are the prefixes (of form a.b.c.d/x) for the four subnets? (Note: All answers should be in dotted decimal notation.)

National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Computer Networks	Course Code:	CS 3001
	Program:	BS (Computer Science)	Semester:	Spring 2026
	Duration:	15 minutes	Total Marks:	15
	Paper Date:		Section	6B1
	Exam Type:	Quiz 4 - Chapter 4	Page(s):	2

Student Name

Roll No.

Section:

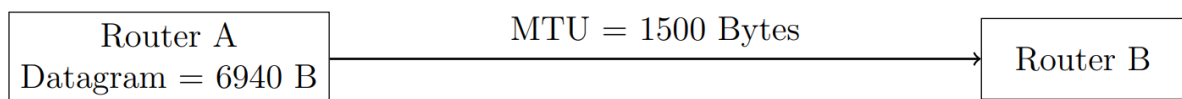
Q1. Carefully read the Question before attempting:

[5 marks]

An IPv4 datagram with a total length of 6940 bytes (20 bytes header + 6920 bytes payload) originates at Router A. It must traverse a link with an MTU of 1500 bytes to reach Router B. Determine the values for the following fields for each fragment generated:

1. Total length of each fragment (header + data).
2. Offset value of each fragment.

Note: The length field in IP is Total Length (Header + Data). The Offset is in units of 8 bytes.



Solution:

- MTU: 1500 Bytes.
- Max Payload per Fragment: $1500 - 20 \text{ (Header)} = 1480 \text{ Bytes}$.
- Total Data Payload: $6940 - 20 = 6920 \text{ Bytes}$.

Fragment #	Total Length (Bytes)	Offset (/8)
1	1500	0
2	1500	185
3	1500	370
4	1500	555
5	1020	740

Note: Offset calculation example for Frag 2: $1480/8 = 185$.

Q2. Carefully read the Question before attempting:

[10 marks]

Consider the router and the two attached subnets (for two organisations) below (A and B). The number of hosts is also shown below. The subnets share the 23 high-order bits of the address space: 192.168.68.0/23.

- **Organisation A has 76 hosts.**
- **Organisation B has 48 hosts.**

Assign subnet addresses to each of the subnets (A and B) so that the amount of address space assigned is minimal, and at the same time leaving the largest possible contiguous address space available for assignment if a new subnet were to be added. Then answer the questions below:

1. What is the subnet address of subnet A? (CIDR notation)
2. What is the broadcast address of subnet A?
3. What is the starting address (first usable) of subnet A?
4. What is the subnet address of subnet B? (CIDR notation)
5. What is the ending address (last usable) of subnet B?

Solution:

1. 192.168.68.0/25
2. 192.168.68.127
3. 192.168.68.1
4. 192.168.68.128/26
5. 192.168.68.190

SOLUTION QUIZ-04 BSE-6A

QUESTION-01

1. **First Address of block:** 172.31.192.0
2. **Total number of addresses:** 8,192
3. **First Useable IP:** 172.31.192.1
4. **Last Useable IP:** 172.31.223.254
5. **Broadcast Address:** 172.31.223.255

QUESTION-02

Total Length (Bytes)	IP Header Length (Bytes)	Identification (ID) (Hexadecimal)	DF Flag	MF Flag	Fragment Offset
596	20	0xABCD	0	1	0
596	20	0xABCD	0	1	72
348	20	0xABCD	0	0	144

SOLUTION QUIZ-04 BSE-6B

QUESTION-01

1. **First Address of block:** 192.168.40.0
2. **Total number of addresses:** 2048
3. **First Useable IP:** 192.168.40.1
4. **Last Useable IP:** 192.168.47.254
5. **Broadcast Address:** 192.168.47.255

QUESTION-02

1.

WAN Side Address (Public)	LAN Side Address (Private)
203.0.113.5:5001	192.168.1.10:3345
203.0.113.5:5002	192.168.1.11:3345

2. Explanation of Distinction When the web server (128.119.245.12) replies, it swaps the source and destination fields from the packet it received.

- **For Host A:** The server sends the reply to 203.0.113.5 on destination port 5001.
- **For Host B:** The server sends the reply to 203.0.113.5 on destination port 5002.

The router examines the **Destination Port** of the incoming packet from the WAN. It looks up this port in the NAT table:

- If the port is **5001**, it rewrites the Destination IP to 192.168.1.10 and Port to 3345.
- If the port is **5002**, it rewrites the Destination IP to 192.168.1.11 and Port to 3345.

**National University of Computer and Emerging Sciences,
Lahore Campus**

	Course Name:	Computer Networks	Course Code:	CS 3001
	Program:	BS (Software Engineering)	Semester:	Spring 2026
	Duration:	15 minutes	Total Marks:	15
	Paper Date:	16-April-2026	Section	6A , 6B
	Exam Type:	Quiz 4 - Chapter 4	Page(s):	2

Student Name

Roll No.

Section:

Q1. Consider an IP address block where one of its address is 172.31.200.130/19. Please provide the following: **[1 * 5 = 5 Marks] [CLO 3]**

First Address of block: _____

Total number of addresses: _____

First Useable IP: _____

Last useable IP: _____

Broadcast Address: _____

Q2. Host A sends a single IPv4 datagram to Host B over a network path that contains a single intermediate router (R1). The original datagram exiting Host A has the following values in its IPv4 header: **[10 Marks] [CLO 3]**

Total Length (Bytes)	IP Header Length (Bytes)	Identification (ID) (Hex)	DF Flag	MF Flag	Fragment Offset
1500	20	0xABCD	0	0	0

The Maximum Transmission Unit (MTU) on the link connecting Host A to Router R1 is 1500 Bytes. The MTU on the link connecting Router R1 to destination Host B is 600 Bytes.

Draw a 6-column table (identical to the one above) and fill in the corresponding header details for *every* fragment created and forwarded by Router R1.

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Student Name

Roll No.

Section:

Q1. Consider an IP address block where one of its address is 192.168.45.77/21. Please provide the following: **[1 * 5 = 5 Marks] [CLO 3]**

First Address of block: _____

Total number of addresses: _____

First Useable IP: _____

Last useable IP: _____

Broadcast Address: _____

Q2. A home router connects a private network (192.168.1.0/24) to the ISP. The router's WAN (Public) IP is 203.0.113.5. Two internal devices establish connections simultaneously:
[6+4 = 10 Marks] [CLO 3]

- **Host A (192.168.1.10)** accesses a web server at 128.119.245.12:80 using source port 3345.
- **Host B (192.168.1.11)** accesses the same web server at 128.119.245.12:80 using source port 3345 (coincidentally the same source port).

Tasks:

1. Construct the **NAT Translation Table** that would exist inside the router to support this scenario. (Create columns for WAN Side and LAN Side).
2. Explain precisely how the router distinguishes the returning packets from the web server, ensuring Host A's traffic doesn't go to Host B.

National University of Computer and Emerging Sciences (Lahore Campus)

Quiz 4: Network Layer (Chapter 4)

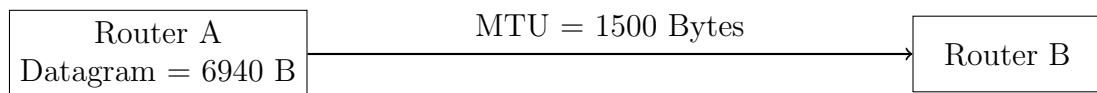
Name: _____ Roll No: _____ Section: BSE-6B1 (Spring 2026)

1. (5 points) **IPv4 Fragmentation**

An IPv4 datagram with a total length of 6940 bytes (20 bytes header + 6920 bytes payload) originates at Router A. It must traverse a link with an MTU of 1500 bytes to reach Router B. Determine the values for the following fields for each fragment generated:

1. Total length of each fragment (header + data).
2. Offset value of each fragment.

Note: The length field in IP is Total Length (Header + Data). The Offset is in units of 8 bytes.



2. (10 points) **Subnet Addressing**

Consider the router and the two attached subnets (for two organisations) below (A and B). The number of hosts is also shown below. The subnets share the 23 high-order bits of the address space: 192.168.68.0/23.

- **Organisation A** has **76** hosts.
- **Organisation B** has **48** hosts.

Assign subnet addresses to each of the subnets (A and B) so that the amount of address space assigned is minimal, and at the same time leaving the largest possible contiguous address space available for assignment if a new subnet were to be added. Then answer the questions below:

1. What is the subnet address of subnet A? (CIDR notation)
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National University of Computer and Emerging Sciences (Lahore Campus)

Quiz 4: Network Layer (Chapter 4)

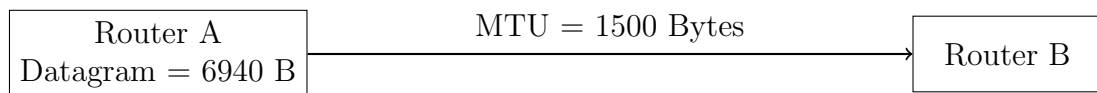
Name: _____ Roll No: _____ Section: BSE-6B1 (Spring 2026)

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1. Total length of each fragment (header + data).
2. Offset value of each fragment.

Note: The length field in IP is Total Length (Header + Data). The Offset is in units of 8 bytes.



Solution:

- **MTU:** 1500 Bytes.
- **Max Payload per Fragment:** $1500 - 20 \text{ (Header)} = 1480 \text{ Bytes}$.
- **Total Data Payload:** $6940 - 20 = 6920 \text{ Bytes}$.

Fragment #	Total Length (Bytes)	Offset (/8)
1	1500	0
2	1500	185
3	1500	370
4	1500	555
5	1020	740

Note: Offset calculation example for Frag 2: $1480/8 = 185$.

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Solution:

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National University of Computer and Emerging Sciences, Lahore Campus

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Section:

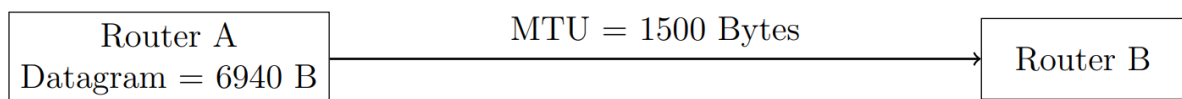
Q1. Carefully read the Question before attempting:

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Q2. Carefully read the Question before attempting:

[10 marks]

Consider the router and the two attached subnets (for two organisations) below (A and B). The number of hosts is also shown below. The subnets share the 23 high-order bits of the address space: 192.168.68.0/23.

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National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Computer Networks	Course Code:	CS 3001
	Program:	BS (Computer Science)	Semester:	Spring 2024
	Duration:	15 minutes	Total Marks:	15
	Paper Date:	2-Apr-2024	Section	6A
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Section:

Q1. Encircle the correct option:

[5 marks] [CLO 1]

1. An IP packet has arrived with the first 8 bits as 0100 0110. Which of the following is correct?
 - a) IPv4, Header Length: 24 bytes
 - b) IPv4, Header Length: 32 bytes
 - c) IPv6, Header Length: 24 bytes
 - d) IPv6, Header Length: 32 bytes
2. In Classless Inter-Domain Routing (CIDR), the network ID in an IPv4 address is:
 - a) always 8 bits long
 - b) always 16 bits long
 - c) always 24 bits long
 - d) none of the above
3. Which IP address(es) should never be assignable to hosts?
 - a) Network IP address
 - b) Broadcast address on the network
 - c) Both a and b
 - d) none of above

True/False:

- a. A DHCP client that has been assigned an address for a period can release the address before the expiration time. [T / F]
- b. VLSM is possible with both Classful and Classless IPv4 addressing schemes. [T / F]

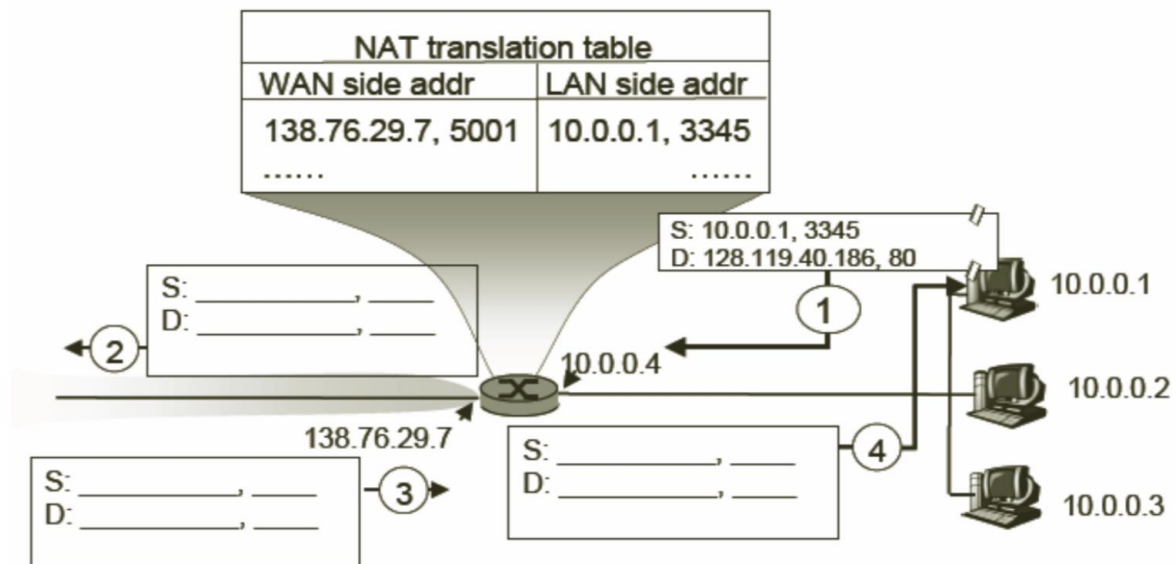
Question 2:

[Marks 10] [CLO 1]

a) The diagram below shows a packet traveling through a NAT router. Packet 1 is sent from the internal host (S) to the NAT router, packet 2 is sent from the NAT router to the external web server (D), packet 3 is received from the web server by the NAT router, and packet 4 is sent by the NAT router to the original host.

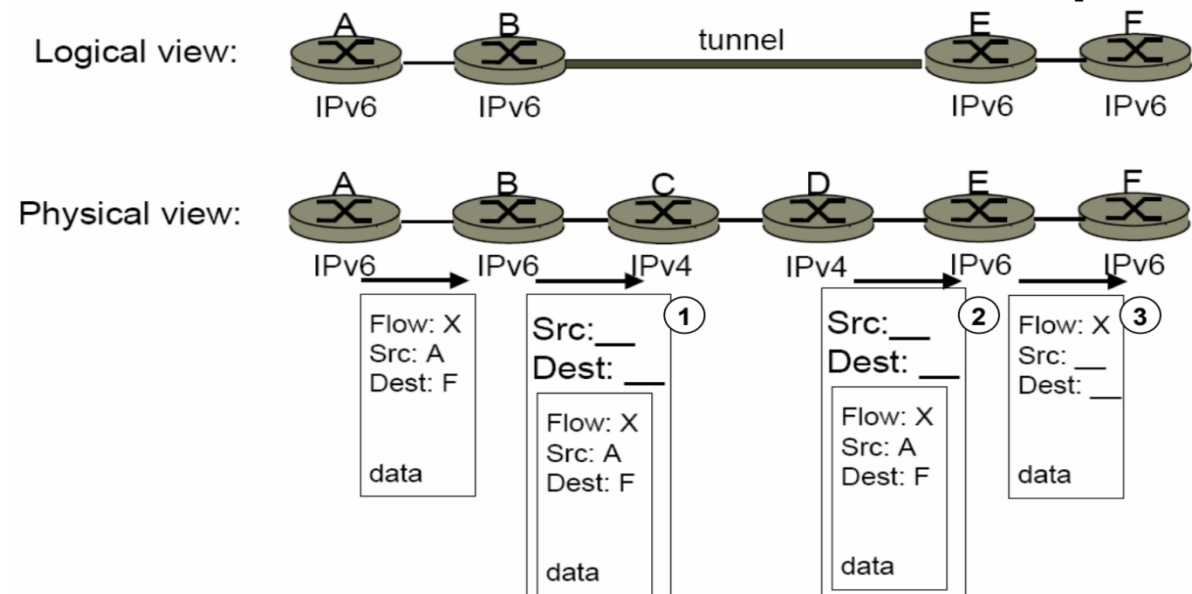
Fill in the missing source and destination IP addresses and port numbers in packets 2-4.

[6 Marks]



b) The diagram below shows an IPv6 packet tunneled over IPv4 across routers with IP addresses A, B, C, D, E & F. Fill in the missing source and destination addresses at places/packets marked 1, 2, and 3.

[3 Marks]



c) Name two modifications of IPv6, from the IPv4, that allow a router to process a packet quicker.

[1 Mark]

SOLUTION QUIZ-04 BCS-6A

QUESTION-01

1. A
2. D
3. C

TRUE/FALSE

1. True
2. False

QUESTION-02

a) SOLUTION:

P2: S=138.76.29.7, 5001 D=128.119.40.186, 80

P3: S=128.119.40.186, 80 D=138.76.29.7, 5001

P4: S=128.119.40.186, 80 D=10.0.0.1, 3345

b) SOLUTION:

1: S=B D=E

2: S=B D=E

3: S=A D=F

c) Any one of these:

1: fixed header length

2: no packet fragmentation at routers

3: no checksum

National University of Computer and Emerging Sciences, Lahore Campus

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Section:

Q1. Encircle the correct option:

[5 marks] [CLO 1]

1. What is the purpose of a subnet mask in IP addressing?
 - a) To identify the network / subnet portion of an IP address
 - b) To encrypt IP traffic
 - c) To define the default gateway
 - d) To allocate dynamic IP addresses
2. Which TCP/IP model layer does DHCP work at?
 - a) Internet
 - b) Session
 - c) Application
 - d) Transport
3. An IP layer packet has arrived with the first 8 bits as 0110 0110. Which of the following is correct?
 - a) IPv4, Header Length: 24 bytes
 - b) IPv4, Header Length: 32 bytes
 - c) IPv6, Header Length: 40 bytes
 - d) IPv6, Header Length: 24 bytes

True/False:

- i. DHCP can provide not only IP addresses but also subnet masks, gateways, and DNS server information [T / F]
- ii. Class A IP addresses have the first octet (8-bits) reserved for the network portion, and the remaining 3 octets for hosts. [T / F]

Question 2

For each of the following IP address ranges, specify the network address, broadcast address, and maximum number of host IPs available. (Show all your work).**[5+5=10 Marks] [CLO 1]**

- a) [192.168.100.0/24](#)

Network Address:

Broadcast Address:

Number of Assignable IP addresses:

- b) [100.10.8.0/22](#)

Network Address:

Broadcast Address:

Number of Assignable IP addresses:

SOLUTION QUIZ-03 BSE-6A

QUESTION-01

- 1) A
- 2) C
- 3) C

TRUE/FALSE

- i. True
- ii. True

QUESTION-02

a)

N/W: 192.168.100.0
B C: 192.168.100.255
No. of IPs: 254

b)

NW: 100.10.8.0
BC: 100.10.11.255
No. of IPs: 1022